nature medicine

Perspective

https://doi.org/10.1038/s41591-025-03742-3

Strategies for shortening tuberculosis therapy

Received: 7 October 2024

Accepted: 25 April 2025

Published online: 13 June 2025

Check for updates

Véronique A. Dartois[1], Valerie Mizrahi[2], Radojka M. Savic[3], Jared A. Silverman[4], David Hermann[5] & Clifton E. Barry III[6]

In the absence of effective patient-stratification approaches, tuberculosis (TB) treatment relies on overtreating most patients to ensure high cure rates. Shortening treatment duration without compromising efficacy is therefore high on the agenda of the global TB community. While new and better drugs are certainly needed, we argue that innovative but rational treatment strategies, using both new and existing therapies, will help achieve this goal. There is growing recognition that patient stratification, based on host and pathogen factors, is key to delivering the right drug regimen for the right duration. In this Perspective, we review the current knowledge on the heterogeneity of TB disease and propose approaches to optimize treatment duration in distinct patient groups, taking into consideration the realities of TB control globally. We emphasize key insights that improve the understanding of bacterial vulnerabilities in patients with easy-to-treat and hard-to-treat TB, helping to reduce diagnostic uncertainties. We explore how the TB research community can integrate disease biology, pathology and symptoms, to rethink therapeutic strategies and reduce TB treatment duration.

The recommended 6-month regimen to treat TB is driven by the duration required to cure the most severe forms of pulmonary TB, while shorter durations appear sufficient to cure less extensive disease[1,2]. Thus, the current standard of care (SOC) can be viewed as a compromise that relies on overtreating most for the benefit of few[1]. This is understandable from a health system and public health perspective in which regimen simplicity and high cure rates are necessary. However, given recent advances in biomarkers of disease heterogeneity, portable technologies to stratify patients and the healthy clinical development pipeline, it is time to revisit and better define the optimal TB treatment strategies across the disease spectrum[1].

Globally, the aim of TB research and development organizations is to discover and design highly effective, markedly shorter (<2 months) pan-TB treatment regimens, where 'pan-TB' is defined as a regimen that can be administered without prior knowledge of the patient's drug resistance profile[3,4]—and coadministration of mechanistically novel agents is expected to slow the inevitable emergence of clinically meaningful drug resistance. In clinical trials, the efficacy of available treatments for drug-susceptible and multidrug-resistant TB is excellent, with favorable outcomes reported in 85–95% of patients. However, in real-world practice, effectiveness is often as low as 65% (ref. 5), with an average of 88% for first-line regimens[6]. This decreased real-world effectiveness is partly due to limitations in implementing drug susceptibility testing, loss to follow-up, suboptimal drug availability and adherence, and differences in patient populations—for example, trial inclusion/exclusion criteria can limit patient heterogeneity and disease severity relative to real-world conditions.

Here, we provide an update on our understanding of disease heterogeneity[4] and how this is informing strategies for patient stratification and TB treatment-shortening approaches. We also discuss the

[1]Center for Discovery and Innovation, Hackensack Meridian School of Medicine, Nutley, NJ, USA. [2]Molecular Mycobacteriology Research Unit, Institute of Infectious Disease and Molecular Medicine & Department of Pathology, University of Cape Town, Cape Town, South Africa. [3]Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA. [4]Gates Medical Research Institute, Cambridge, MA, USA. [5]Gates Foundation, Seattle, WA, USA. [6]Tuberculosis Research Section, Laboratory of Clinical Immunology and Microbiology, National Institutes of Health, Bethesda, MD, USA. e-mail: veronique.dartois@hmh-cdi.org; cbarry@niaid.nih.gov

formidable challenge of dealing with this most complex infectious disease while requiring standardized and simple treatment guidelines that can be implemented in all settings, for effective disease control around the world.

Clinical TB is heterogeneous

Over the past 15 years, it has been increasingly recognized that TB presents as a continuum of manifestations rather than a bimodal distribution between active and latent TB[7]. TB disease is more usefully described as a spectrum of responses to *Mycobacterium tuberculosis* (Mtb) infection, with clear implications for preventive therapy and treatment-shortening strategies[1,8]. Disease trajectories diverge from the point of infection. Some take an accelerated course, while others may take a more chronic or undulating course and may self-clear or self-heal with or without the development of symptoms or an adaptive immune response[9–11]. Active symptomatic disease itself is profoundly heterogeneous.

The drivers of this heterogeneity are poorly understood and therefore individual trajectories remain largely unpredictable[12]. Global trends point to geographical differences in disease progression and severity[13], which may be related to health-seeking behavior (delayed presentation results in increased disease extent), Mtb lineage[14–16], and may also be a consequence of host–pathogen coevolution[17,18]. For instance, the occurrence of severe cavitary TB (characterized by hollow spaces filled with necrotic material, gas and Mtb bacteria surrounded by a discrete fibrotic wall) is overall higher in African than in Asian populations[19], and lineage 1 was recently associated with higher rates of cavitary TB and mortality[16]. Patients with comorbidities also follow differential trajectories, with unique features and disease progression profiles. HIV infection is the strongest known risk factor for progressing from latent to active TB disease. Likewise, the relative odds of developing TB in people with diabetes compared to those without diabetes ranges from 2.4 to 8.3 (ref. 20). Finally, infants, young children, primary school children and adolescents each present markedly different rates of TB incidence, disease presentation and patterns of disease progression[21].

Considerations relating to the site of disease have recently begun to influence drug regimen design in clinical trials. TB meningitis is the most devastating and difficult-to-treat form of the disease, with poor treatment response, high mortality and morbidity, and lifelong neurological sequelae (in the form of cognitive impairment, motor deficit and optic atrophy) occurring in a large fraction of survivors. Treatment recommendations call for the current pulmonary SOC and an extended 10-month treatment continuation phase in adults[22], but with increasing awareness that these regimens require substantial overhaul to reach therapeutic concentrations in cerebral sites of infection, including spinal compartments[23–25]. At the other end of the severity spectrum, intrathoracic lymph node TB without airway obstruction or peripheral lymphadenopathy, and TB pleural effusion without pulmonary TB are often paucibacillary and sputum culture negative, and may be cured with simpler or shorter regimens than the current SOC[26].

Classification systems for individuals with TB disease, such as the American Thoracic Society's TB 0–TB 5 (ref. 27), have been used by public health programs. This classification distinguishes between exposure only, non-clinically active and clinically active TB but does not consider active disease severity. A recent promising development in defining the spectrum of TB disease is the conceptual mapping of eight TB states, based on systematic literature mining, to standardize scientific communication and inform clinical research and public health practice[11,28]. Of these eight states, the first three—infection eliminated by the innate (state 0) or acquired (state I) immune response, or infection not eliminated but controlled by the immune system (state II)—cannot be distinguished with existing diagnostic tools, but their study can influence future disease control strategies. States IV to VI are relevant to disease classification for the purpose of curative therapy. State IV is bacteriologically positive without symptoms, often described as subclinical (or asymptomatic) disease; state V includes signs or symptoms associated with TB or active TB; and state VI includes severe or disseminated TB disease, differentiated from the previous state through terms such as cavitary or extensive disease.

Subclinical or incipient TB is increasingly recognized as a disease state that must be diagnosed and treated more effectively to preclude further transmission[11,13,29]. Although these patients do not experience clinical symptoms, microbiological and radiographical characteristics vary widely, from negative sputum culture and absence of radiographic or positron-emission tomography (PET)/computed tomography (CT) findings to sputum positivity and cavities present in a small fraction of people[30,31]. Thus, subclinical TB comes with a complex array of bacterial burden and radiographic abnormalities, which presents considerable diagnostic challenges[32] and impacts disease trajectories[33] and therapeutic strategies. The microbiological complexity is further compounded by the presence of viable Mtb in respiratory aerosols irrespective of sputum culture status and chemotherapy[34]. In addition, the time at which each individual will seek care along their disease trajectory, or whether they will seek care at all (regardless of clinical symptoms), is influenced by sociocultural and other factors, further highlighting the critical role of active case finding.

Clinical phenotypes inform the required treatment duration

In middle- to high-resource countries, clinical practice already adjusts the treatment duration for drug-susceptible TB on the basis of disease severity, site of disease and patient profile. For example, the US Centers for Disease Control and American Thoracic Society guidelines recommend 4 months of treatment for chest X-ray-negative and smear-negative but symptomatic disease, 9 months for HIV co-infected persons, and a 3-month extension of the continuation phase for people with large cavities at baseline and those who remain sputum culture positive after 2 months of treatment[35]. In low-resource countries, World Health Organization (WHO) guidelines are more typically followed, which recommend 6 months of treatment for all new TB cases[36], still calling for extended treatment for infections of the central nervous system, bone or joint, and 4 months of treatment for children/adolescents with 'non-severe' TB[22].

The history of TB regimen development that led to the current SOC has been well reviewed elsewhere[37]. In brief, treatment duration was exhaustively explored during the development of the current four-drug combination therapy; the pattern of relapse after drug discontinuation (for the collection of agents available at the time) was distinctly a steep, monotonic function of treatment duration (Fig. 1). The distinct shape of the treatment duration–response curve likely reflects a range of disease severities in which most patients are cured with 3 months of therapy, but a subset only respond after 5–6 months of treatment. Also noteworthy is that even the extension of treatment to 9 months with the same regimens did not prevent relapse in 1–2% of patients. Overall, this suggests that there are three basic strata of patients with TB: 'easy to treat' (ETT; 80%), 'hard to treat' (HTT; 15–18%) and 'extremely hard to treat' (1–2%)[19].

More recent clinical trials support this proposition. Three large phase 3 trials examined the possibility that adding a fluoroquinolone to standard treatment could reduce the duration of therapy to 4 months[38–40]. All three trials failed to demonstrate non-inferiority of the shortened regimen when comparing relapses across all patients. A reanalysis of the individual patient data subsequently showed that the 4-month regimens were non-inferior in a large proportion of patients with minimal disease but were clearly inferior in patients with more extensive cavitary disease[2].

Patients with HTT disease tend to have specific pathological features (cavitary disease in particular), and multiple studies have shown that such patients have a higher risk of disease relapse even

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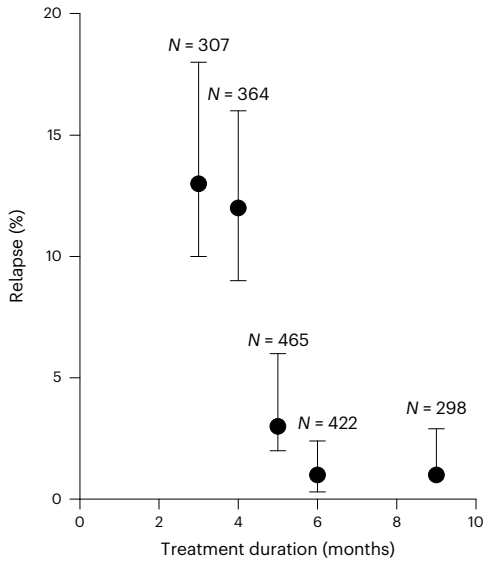

Fig. 1 | Microbiological relapse rates as a function of treatment duration. The graph shows pooled results with 95% CIs and the total number of participants per trial, plotted from historical data in ref. 107. Regimens contained various combinations of isoniazid, rifampicin, pyrazinamide, ethambutol and streptomycin. It was recognized from this analysis that more than 80% of all patients with TB were cured within 3 months of combination therapy, supporting the notion that we overtreat a large fraction of patients with the current SOC. For the remaining 20% of patients, it wasn't until >5 months of therapy that relapse rates could be driven down to less than 5%, which supported 6 months as the shortest duration providing near-maximal efficacy across a heterogeneous population.

after 6 months of chemotherapy[41–44]. Repeated PET/CT exams in the recent PredictTB trial showed that baseline cavities, particularly those with thick walls and still containing the necrotic material known as caseum (so-called 'dirty cavities'), are the actual site of disease relapse in patients who received 4 months of treatment[19], confirming that the difference in treatment response between ETT and HTT disease centers around cavities. We posit that optimized drug regimens must be tailored to disease type: drug combinations that shorten treatment duration of cavitary disease may not shorten treatment of ETT disease, in which the predominant immunopathological features are inflamed airways and 'clean' (devoid of caseum) cavities, and vice versa (Fig. 2). The notion that combining rapidly bactericidal agents selected in preclinical TB models may deliver dramatic treatment shortening for individuals with ETT disease but not for those with HTT disease is being actively investigated.

Different bacterial phenotypes may require different treatments

TB lesions from 1,500 autopsies were meticulously characterized by Georges Canetti in the 1940s. He was the first to investigate the heterogeneity of both the histopathology and the 'histobacteriology' of these lesions[45]. Canetti first described primarily cellular lesions devoid of necrosis, in which small numbers of bacilli were predominantly in macrophages. As caseation occurred (whereby necrosis results in a 'cheese-like' texture of the tissue or caseum) in a subset of lesions, larger numbers of bacilli were noted in the necrotic foci. As caseous lesions matured, they became paucibacillary with only two to five bacilli detected in 100 fields, but the caseum was not sterile, since injecting it to Guinea pigs resulted in disease. The remaining bacilli in these closed lesions are poised to explosively multiply after liquefaction of the caseum. Liquefaction of caseum produces a cavity, defined by the 'capsule' (tissue resembling sclerosis) and an interior necrotic zone of variable extent, being drained with its bacillary contents into the airways. This interior necrotic zone was always associated with bacilli too numerous to count (a fact first observed by Robert Koch when he discovered the tubercle bacillus)[46]. Canetti also described another type of lesion, referred to as 'polymorphonuclear alveolitis', which was dominated by neutrophils and contained high numbers of bacilli.

Consideration of whether cavities are 'open' (directly connected to airways) or 'closed' (cut off from airways, more commonly designated as a necrotic lesion), has a poorly understood impact on the metabolic state of the bacteria. In open cavities resected from individuals with TB, plating the contents onto standard growth media produces colonies after the standard incubation time. However, the plated contents of many closed cavities produced colonies only after prolonged incubation for 3–10 months[47]. Understanding these differentially culturable bacilli in sputum is an ongoing focus of many labs (reviewed in ref. 48).

Modern imaging methods such as micro-CT combined with PET have been used to derive three-dimensional reconstructions of complex branched structures with often subtle connections to the airways and elucidate the dynamics of lesions over time[49,50]. These studies revealed that lesions can emerge from active cavities through the

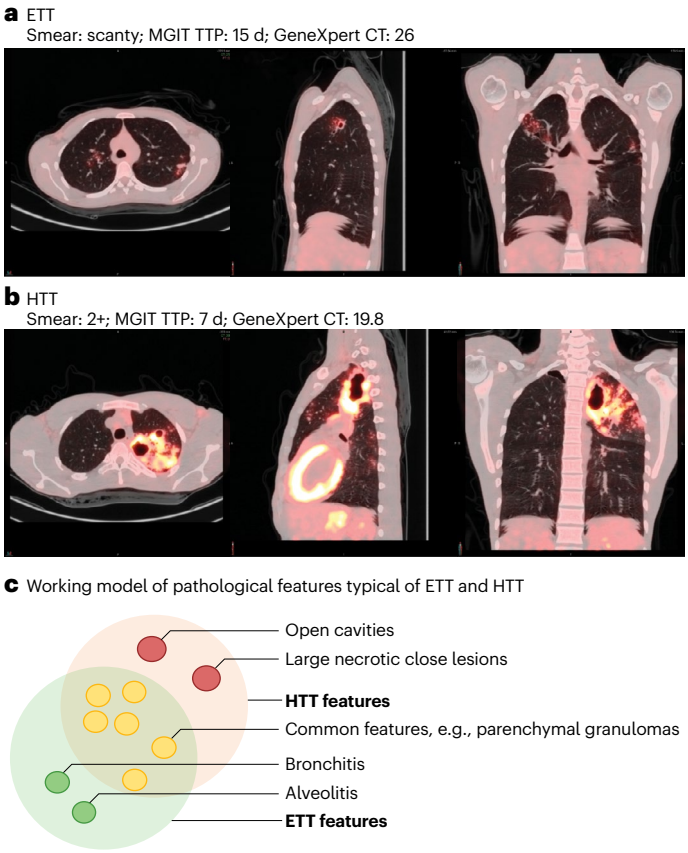

Fig. 2 | Immunopathological features of typical ETT and HTT TB. a,**b**, Two patients treated for 4 months in the PredictTB trial, one successfully (**a**) and one not (**b**). ETT patients typically show thin-walled cavities and pulmonary nodules, visible in both the left and right lungs of the patient shown in **a**. HTT patients typically show large, thick-walled cavities as shown in the left upper lobe in **b** which was also the site of disease relapse 6 months after this patient discontinued treatment. In the initial sputum nucleic acid amplification test (GeneXpert), patient A showed a cycle threshold (Ct) value of 26.1 indicative of low sputum burden, while patient B showed a Ct of 19.8 indicative of high sputum burden[19]. **c**, Working model illustrating hypothetical pathological features that may create a bottleneck to cure in patients with ETT TB and those with HTT TB and how they may overlap. Parenchymal granulomas refer to small cellular lesions. Alveolitis refers to neutrophil-driven polymorphonuclear airway disease in terminal airways and bronchitis in larger airways, the latter resulting from bronchial spread of caseous spills.

Table 1 | Randomized controlled trials evaluating shorter-than-standard treatment duration in participants with limited disease

Indication	Duration	Trial acronym/ sponsor	Drug(s)	Primary endpoint/ outcome[a]	Study identifier or reference
Completed					
Non-severe pediatric DS-TB (HIV+ and HIV−)	4 mo versus 6 mo	SHINE (phase 3)	2HRZ (+/−E)/4HR (6 mo) 2HRZ (+/−E)/2HR (4 mo)	No difference in the effect of randomized treatment duration on treatment outcomes by HIV status: 6 mo: unfavorable outcome = 2% in non-HIV (n=540), and 11% in HIV (n=62) 4 mo: unfavorable outcome = 2% in non-HIV (n=537), and 7% in HIV (n=65)	26
DS-TB Non-cavitary HIV−Smear− / culture + Culture— at 2 mo	4 mo versus 6 mo	Tuberculosis Research Unit, Case Western Reserve University	2HRZE/4HR (6 mo) 2HRZE/2HR (4 mo)	Bacteriological or clinical relapse over 30 mo: 6 mo: 1.6% (n=185) 4 mo: 7% (n=185; study stopped early)	97
DS-TB Smear−/ culture +	4 mo versus 6 mo	Hong Kong Chest Service	$6SHRZ_3$ (6 mo) $4SHRZ_3$ (4 mo)	Bacteriological, clinical or radiological relapse over 60 mo: 6 mo: 5% (n=296) 4 mo: 2% (n=293)	98
DS-TB Smear−/ culture +	4 mo versus 6 mo	Singapore Alexandra Hospital	$2HRZ/4H_3R_3$ (6 mo) 2HRZ/2HR (4 mo)	Bacteriological relapse over 60 mo: 6 mo: 2% (n=53) 4 mo: 0% (n=57)	99
Ongoing					
DS-TB	4 mo	PredictTB (phase 2)	HRZE	Change in radiographic characteristics from baseline to one month, and markers of residual bacterial load at the end of treatment, to identify patients who are cured.	NCT02821832
XDR-TB	6 mo versus 9 mo	endTB-Q (phase 3)	BeDeCLi versus WHO-recommended treatment	Randomized, controlled, non-inferiority open-label trial evaluating efficacy according to extent-of-disease phenotype.	NCT03896685
DS-TB	4 mo	PRESCIENT (phase 2c)	BCZD versus SOC (HRZE)	Open-label, randomized study (stratification by presence of lung cavitation and HIV status). The primary analysis will focus on time to liquid culture conversion at week 8 in the BCZD arm versus HRZE.	NCT05556746

BeDeCLi, bedaquiline, delamanid, clofazimine and linezolid; BCZD, bedaquiline, clofazimine, pyrazinamide and delamanid. DS-TB, drug-sensitive TB; DR-TB, drug-resistant TB; XDR-TB, extensively resistant TB; 2HRZ (+/−E), isoniazid, rifampicin and pyrazinamide (with or without ethambutol (E)) for 2 months; mo, months; 4HR or 2HR: isoniazid and rifampicin for 4 or 2 months; $6SHRZ_3$ or $4SHRZ_3$, streptomycin, isoniazid and rifampicin daily and pyrazinamide thrice weekly for 6 months or 4 months; $4H_3R_3$, isoniazid and rifampicin thrice weekly for 4 months.
[a]Outcome described for completed trials only.

bronchial spread of cavity caseum, perhaps giving rise to bronchial inflammation—similar to Canetti's 'polymorphonuclear alveolitis', in additional sites[50]. The features of these lesions imply a wide range of physiological states of the bacilli contained within[51]: intracellular in different immune cell types, extracellular in the vicinity of these cells and in non-vascularized caseum, actively replicating and nonreplicating in a 'persisting' metabolically quiescent state. These populations are known to display differential susceptibilities to individual drugs, and since they are differentially abundant or present in different individuals, optimizing drug regimens as a function of disease phenotype is expected to deliver better cure rates and shorten treatment duration[48].

Drug resistance and new TB drug regimens

The stunning success of the bedaquiline–pretomanid–linezolid combination in the Nix-TB trial[52] demonstrated that >90% cure rates can be achieved against multidrug-resistant TB and that new regimens can dramatically impact treatment duration and regimen simplicity. However, resistance to bedaquiline and pretomanid was detected soon after their launch. Resistance to these agents can arise during treatment, resistant organisms can be transmitted from patient to patient, and resistance is increasing in frequency in some settings[53,54]—all of which highlights the need for preemptive strategies that anticipate and minimize opportunities for resistance to develop.

The evaluation of streptomycin TB monotherapy was among the first randomized controlled trials ever conducted and despite radiological improvement and a notable reduction in deaths, treatment resistance emerged in 80% of the patients[37]. Like most bacterial pathogens, Mtb has acquired resistance to all drugs it has been routinely exposed to[55], including those only recently introduced into clinical practice (bedaquiline[56], delamanid and pretomanid[57])—even when used in combination with two or three other agents. The observation that *mmpR5* resistance mutations predate the introduction of bedaquiline[58] is helping the community draw lessons from past mistakes and develop drug susceptibility companion tests to be used in clinical trials and co-launched with future novel drugs. Deployment of such companion tests is critical because clinical resistance rates are not always consistently predicted by in vitro studies, as exemplified by isoniazid and rifampicin. In vitro, resistance acquisition rates are much higher for isoniazid than rifampicin[55], but acquired monoresistance rates in patients on multidrug therapy are 0.3% for both isoniazid (95% confidence interval (CI), 0.06% to 2.7%) and rifampicin (95% CI, 0% to 0.9%) in comprehensive meta-analyses[59,60]. Thus, the development of a new drug or regimen should be accompanied by the development of drug susceptibility testing technologies along with the introduction of population-level surveillance programs for drug susceptibility.

How can resistance emerge rapidly in the context of three-drug or four-drug therapy, as recently seen for bedaquiline, delamanid

Table 2 | Range of plausible treatment strategies for pulmonary TB[a]

Treatment strategy	Strengths	Challenges	Knowledge gaps and unmet needs
One size fits all: New short regimen equally effective in rifampicin-resistant and susceptible populations, including asymptomatic sputum-positive patients.	• Simplest: No DST to initiate treatment. • Potential for improved completion rates, outcomes and maximal impact on incidence reduction.	• Even shorter regimens risk overtreating a proportion of patients. • Emergence of clinically relevant resistance requiring DST. • Shortening to 2 months or less is required to overcome implementation hurdles and improve adherence.[b]	Designing regimens to protect from resistance emergence and optimizing PK–PD across all sites of infection.
Stratification by rifampicin resistance status (current SOC)	• Simplicity. • Minimizes risk of undertreating.	• DST poorly implemented in low-resource settings. • Low prescription completion rates. Overtreats a sizable fraction of patients risking side effects and patient well-being. • Adds cost.	Improved (faster, cheaper, easier) DST to guide therapy.
One regimen, varying durations based on patient factors	• First step toward minimizing the fraction of overtreated and undertreated patients. • Lowest barrier to implementation worldwide; short-term feasibility.	• Incremental improvement on current SOC strategy. • Can be implemented in high-resource settings only[35]; implementation challenging for community-based DOT. • Added complexity of stratification through digital chest X-ray or other tests must be cost effective in low-resource settings[100,101].	Improved (faster, cheaper, easier) biomarkers to guide therapy.
One regimen, varying doses for drug-susceptible and drug-resistant and patient factors	• Modifies SOC to minimize the fraction of overtreated and undertreated patients. • Low barrier to implementation worldwide; short-term feasibility.	• Few drugs are eligible for higher-than-standard doses. • DST poorly implemented in low-resource settings.	Clinical trials to assess the efficacy of modified-dose regimens across disease severity spectrum. Routine implementation of therapeutic drug monitoring and safety monitoring. Accessible, geographically diverse DST/MICs.
Distinct regimens for ETT and HTT disease	• Maximizes trade-off between precision and simplicity. • Most effective approach if drugs that shorten treatment in HTT do not shorten treatment in ETT (Fig. 2c).	• Evidence that patients with ETT disease and those with HTT disease need different regimens is lacking. Biomarkers that safely define ETT versus HTT have not been identified yet. Clinical research and trials required, thus not a short-term solution.	Development and validation of a globally accepted—deliverable, scalable and affordable—biomarker that allows patient stratification. Uncovering novel drug mechanisms. Systematic exploration of pharmacodynamic interactions for regimen optimization.
Regimen cycling (different regimens for intensive and continuation phases)	• Supported by NHP observations that regimen effectiveness varies over time. Antibiotic cycling is an actively pursued strategy against other bacterial infections.	• Lack of data on how, when and where this approach may work. • Changing drugs may require additional DST, safety monitoring and cross-regimen drug–drug interaction studies. • Potentially complicates compliance and supply chain.	Extensive preclinical and clinical research required to demonstrate value and optimize practice.
Personalized therapy (Fig. 3)	• Maximizes precision: optimal drug regimen, doses and duration for each patient.	• Most complex to implement. • May require advances in technology. • As regimens become shorter, the benefit to patients and health care systems may bear diminishing returns.	Development and validation of a globally accepted biomarker that allows patient stratification. Digital patient monitoring to assess treatment response and pharmacogenomics to enable stratified care.

DST, drug susceptibility testing; DOT, directly observed therapy; NHP, nonhuman primate. [a]While important and deserving attention, TB meningitis and other disease manifestations require separate regimen optimization. [b]The new and successful 4-month regimen for drug-susceptible TB (rifapentine–moxifloxacin–isoniazid–pyrazinamide[36]) is seldom implemented in low-resource settings for that reason and due to the lack of a fixed-dose combination.

and pretomanid, seemingly faster than for the drug regimens first used in the 1960s and 1970s? First, these drugs are collectively linked to a large panel of resistance mutations: *ddn*, *fbiA*, *fbiB*, *fbiC*, *fbiD* and *fgd1* for delamanid and pretomanid, and *atpE*, *Rv0678* and *pepQ* for bedaquiline. Second, bedaquiline resistance due to drug efflux by the transporter MmpL5 (upregulated by mutations in the *Rv0678* gene) was identified in geographically diverse Mtb genomes before its launch, attributed to either historic clofazimine use or adaptation to other environmental stressors[55]. Third, the frequency of resistance observed in vitro does not reliably predict acquired resistance in vivo, due to differential fitness costs (whereby resistant bacteria tend to be less 'fit' than their drug-sensitive counterparts). This highlights the challenges and pitfalls associated with identifying an acceptable threshold of resistance frequency in vitro to support the approval of a novel agent, be it alone or in combination. Only long-term studies assessing epidemiological and fitness costs of resistance mutations across patient populations can provide compelling answers.

In addition to these factors, de facto monotherapy at the site of infection (owing to all but one agent in the drug combination not effectively reaching the site) may drive acquired resistance. The caseous foci of necrotic lesions and cavities, which constitute treatment bottlenecks, are devoid of vascular supply and thus rely on passive equilibration of unbound drug concentrations at steady state, which can take several weeks for highly bound drugs. In addition, the very long terminal half-life of bedaquiline relative to other drugs in combination regimens creates de facto monotherapy for weeks to months following

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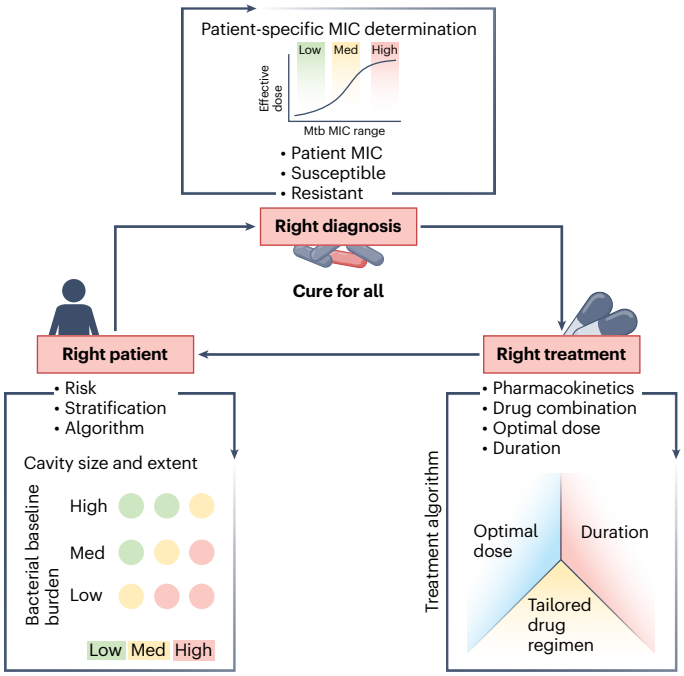

Fig. 3 | Patient-stratification approach. The figure illustrates the interplay between diagnosis, host–bacterium features and treatment optimization for TB. As a long-term vision, algorithms that integrate disease severity and bacterial burden, comprehensive drug susceptibility profiles, pharmacogenomics, pharmacokinetics and drug–drug interactions will be developed and validated to deliver optimized, patient-tailored drug doses and regimens. Real-time monitoring of treatment response will enable adjustments of treatment duration to avoid overtreating as much as undertreating.

treatment completion. Combined with the profound drug tolerance of nonreplicating, persisting Mtb in caseum[61], these phenomena create spatial and temporal reservoirs of surviving bacteria that may fuel acquisition of resistance after bronchogenic spread and resumption of growth. Adding a fast-acting bactericidal agent during initial treatment would increase the number of effective drugs during this period and help protect from the development of resistance. Approaches that include high-dose isoniazid, novel oral beta-lactams and second-line injectables have been proposed[54].

TB is treated with multiple agents to reduce the emergence of drug resistance, but this simple idea must be tempered by the concept of pleiotropy (whereby one mutation affects two or more phenotypic traits). Acquired or intrinsic resistance to one drug may also enable cross-resistance to other drugs (within or across antibiotic classes), or could result in 'collateral sensitivity', whereby a mutation or pathway that confers resistance to one antibiotic increases susceptibility to another[62–64]. In Mtb, glutamine synthetase has been identified as a collateral metabolic vulnerability elicited by bedaquiline[65]. In mycobacteria, the transcriptional regulator WhiB7 coordinates an adaptive stress response to ribosome stalling, leading to multidrug intrinsic resistance and tolerance, which has important implications for the design of combinations that avoid WhiB7 induction[66–68]. Combination chemotherapy strategies need to incorporate these concepts early in preclinical development—not only to create more durable antibiotic pairings[69], but also to reduce treatment duration by circumventing intrinsic pathways of drug tolerance. Functional genomics and chemical–genetic interactions identified by CRISPR interference[70] are being exploited to identify pathways critical for fitness of resistant mutants[71], suggesting rational approaches to induce sensitivity. These approaches can inform the development of mechanism-based combinations that shorten treatment and delay the emergence of resistance.

Emerging research and development strategies

As noted earlier, 80% of patients with TB likely fall into an ETT category—suggesting a broad benefit from patient stratification if it can be easily incorporated into practice, considering the challenges associated with identification of those eligible for safe treatment shortening, and trials aimed at assessing this using various stratifications are ongoing[1] (Table 1). Initially, patient stratification could be accomplished with a conceptually simple distinction between ETT and HTT disease phenotypes.

Typical PET/CT scans and sputum burden profiles of representative individuals with ETT disease and those with HTT disease are shown in Fig. 2. Since increased precision comes at the cost of simplicity and feasibility, stratification and treatment algorithms may need to be adapted to clinical settings according to resource availability, or be launched at a slower pace in low-resource settings. For instance, a composite signature classifies HTT disease on the basis of cavitation by HIV status, sex, body mass index and smear grade in the absence of chest X-ray[2]. These host factors are consistent with the guidelines from the US Centers for Disease Control and Prevention and the American Thoracic Society[35], and can be collected in most clinical settings. A molecular biomarker signature captured by a regulatory-approved device is a longer-term prospect and would likely be launched gradually around the world. Table 2 describes a range of plausible stratified treatment strategies, with their associated benefits, limitations, knowledge gaps and implementation challenges. Our hypotheses for treatment shortening echo the long-standing principles of clinical pharmacology: get the right drugs to the right patients at the right doses, and for the right duration (Fig. 3). Below, we outline strategies that can be envisioned to tailor drug, dose and duration to patient needs and disease phenotypes[72].

Develop regimens to target biological and spatial characteristics of ETT and HTT disease

Spatial and temporal within-host lesion diversity is the hallmark of Mtb infection and TB[19,73,74]. In these lesions, bacteria exist in various physiological and drug tolerance states, governed by intracellular and extracellular microenvironments. That new lesions appear, and cavities can form even under multidrug therapy[73], reminds us that current drug regimens do not effectively suppress bacterial multiplication at all sites of infection. Retrospective analyses of recent treatment-shortening phase 3 trials indicate that regimens penetrating cavities at concentrations needed to kill drug-tolerant Mtb[61] were successful across ETT and HTT populations. For instance, the first successful 4-month regimen tested in Study 31 (ref. 36) contains two drugs (high-dose rifapentine and moxifloxacin) that achieve excellent cavity caseum coverage—meaning that they reach the concentrations required to kill nonreplicating caseum persisters for the entire dosing interval. We hypothesize that this site must be covered at adequate concentrations for resistance prevention and relapse-free cure of all patients. Retrospective analyses of large trials using simulations of drug regimen coverage in lung lesions are ongoing to confirm this hypothesis[75].

To achieve relapse-free cure (no relapse within 18 months following therapy completion) while shortening treatment duration, a key priority is the identification of vulnerable bacterial targets in subpopulations that constitute treatment 'bottlenecks' in ETT and HTT disease. Preliminary findings[19] point to drug-tolerant, nonreplicating persisters in cavity caseum as the recalcitrant Mtb population that resists therapy and constitutes a source of relapse in HTT disease. To improve sterilization of these bacteria, we can infer vulnerable targets and pathways from successful treatment-shortening trials, bactericidal assays in caseum ex vivo[76] and efficacy studies in nonhuman primates (with longitudinal PET/CT scans as a readout).

While there is clarity on the site of disease that drives outcome and is the site of relapse in HTT ('dirty' cavities filled with caseum[19]), there remains uncertainty regarding the dominant site that drives relapse in ETT. Factors associated with pathogen biology, target vulnerability and

Table 3 | Working hypotheses, biological and clinical uncertainties, and proposed approaches to better target patients with ETT and HTT disease

Hypothesis	Uncertainties	Approaches to address uncertainties
Treatment shortening will be achieved by ensuring that ≥2 regimen components deliver relevant exposures to caseous lesions.	• Optimization of PK–PD coverage in 'dirty' cavities • Metabolic vulnerabilities of MTB in caseum • Role of 'persister' bacteria	• Assessments of lesion/caseum PK–PD in established animal models and lung resected specimens • Ex vivo testing platforms with discovery stage candidates • Mathematical modeling integrating multidrug PK–PD metrics, bacterial regrowth kinetics after treatment[102] and dose optimization to achieve desired exposures in dirty cavities • Integrate RS ratio (a measure of newly synthesized RNA)[103] in PK–PD computational pipelines, and preclinical models • CRISPRi knockdowns in relevant animal and ex vivo models • Investigation of TB metabolic status in NHP models (drug naive and drug treated) with dirty cavities
Treatment shortening will be achieved by ensuring regimens target MOAs associated with the predominant pathologies and Mtb phenotypes in patients with ETT and HTT disease.	• Bacterial populations that constitute treatment bottlenecks in ETT and HTT • Metabolic vulnerabilities of MTB in varying infection states • Role of differentially culturable Mtb in treatment response of patients with ETT and HTT disease	• Assess target vulnerability across growth conditions in vitro and ex vivo • Identify mechanisms of action with the best potential for sterilizing HTT and ETT disease • Detection and enumeration of differentially detectable[104,105] or culturable Mtb[106] before and after treatment of patients with ETT and HTT disease • Clinical trials to ascertain the impact of regimens optimized for patients with ETT and HTT disease integrating utility of RS ratio[99] in predicting treatment shortening
Treatment shortening will be achieved by ensuring all regimen components are dosed optimally.	• Balance between dose increase and safety/tolerability. • Potential need for localized region-specific dosing based on resistance patterns	• Establish MIC distributions as a function of lineages and geographic locations[16] • Systematically include wider dose ranges in phase 2 trials • Characterize the probability of target attainment relative to the distribution of clinical isolate MICs and SOD
Treatment shortening can be achieved by distinguishing two categories of patients for treatment allocation: ETT and HTT.	• Ability to distinguish patients with ETT disease and patients HTT disease	• Practical point-of-care diagnostic using easily measured patient factors (for example, body mass index, wide audio spectrum e-stethoscopes as X-ray substitute) • Radiomic analysis of PET/CT scans and machine learning to develop classifiers for use in stratifying in clinical trials • Biomarker analysis of cohorts of patients with ETT and HTT disease to develop point-of-care routine diagnostics
Treatment shortening will be achieved through the availability of a biomarker(s) of relapse-free cure.	• A validated, simple, low-cost highly predictive biomarker/test of relapse-free cure measured during or at the end of treatment	• Prospectively and longitudinally collect relevant patient characteristics and biospecimens for measurement of exploratory biomarkers • Explore relationship between caseum and sputum Mtb for biomarkers of treatment response • Collect patient covariates that will contribute to modeling/estimation of the appropriate treatment duration for different patient phenotypes (ETT or HTT) • Incorporate relevant findings into drug regimen labels and guidelines enabling a stratified approach where possible

location of dominant bacterial populations in ETT versus HTT disease may require that optimized regimens for each of these features a subset of drugs with different target product profiles. Two likely scenarios must be considered. In the first scenario, treatment-relevant pathological features of ETT are all present in HTT and, therefore, ETT could be treated with the same regimen as HTT, either for shorter durations or with a subset of HTT drugs. In the second scenario, there exist ETT-specific pathological feature(s), and this may constitute a distinct barrier to cure—in which case drugs with ETT-specific mechanism(s) of action and site distribution profiles are required (Fig. 2c). Longitudinal PET/CT imaging of patients on treatment[19] suggests that neutrophil-associated airway disease (described as polymorphonuclear alveolitis by Canetti[45]) harbors one of the dominant bacterial populations in patients with ETT disease. Histopathology analyses have revealed intracellular and extracellular bacilli in and around neutrophils. Together with intracellular bacilli present in pulmonary nodules—such as those within lipid-laden foamy macrophages, where metabolically adapted Mtb resists treatment[77–80]—neutrophil-driven pathology may constitute a cure-limiting ETT burden while also being present in HTT disease. Further mining of data from ongoing and recently completed trials should be undertaken, followed by application of computational pipelines for testing hypotheses of regimen ranking versus outcome.

These questions and approaches to address them are an intrinsic part of a coherent strategy to enable treatment shortening in all patients by simultaneously defining the microbiology and immunopathology of ETT and HTT (and maybe even extremely HTT disease) and developing regimens that target the distinct biology and vulnerabilities of each.

Improve dose selection

Not all TB drugs can be safely administered at higher-than-standard doses, but key drugs with sterilizing potential, such as rifampicin, rifapentine and moxifloxacin, offer scope (albeit modest) for dose increase. High-dose rifampicin (twice the standard dose) is generally well tolerated[81,82] and has been proposed for cavitary TB with high sputum burden[83], after multiple studies showed that 600 mg lies in the steep part of the exposure–response curve[84]. High-dose rifapentine is part of the first successful 4-month regimen for drug-sensitive TB[36]. High-dose moxifloxacin (double dose once daily or standard dose twice daily) is expected to provide superior efficacy on the basis of site-of-disease pharmacokinetic–pharmacodynamic (PK–PD) modeling (with a predicted safety advantage for twice daily)[85] and is being tested in a clinical trial (NCT02409290). It is generally recognized that higher moxifloxacin doses could be safely administered if adequate monitoring is undertaken[86]. Given the rapid emergence of bedaquiline resistance in certain geographies, doses adjusted to local minimum inhibitory concentration (MIC) distributions will be evaluated during the development and implementation of the next-generation diarylquinolines, TBAJ876 and TBAJ587. In clinical trials of new drug candidates and clinical studies with approved regimens, quantifying the relationship between individual drug exposure, drug susceptibility and clinical outcome should deliver an algorithm that quantifies the probability of target attainment for multiple drugs, including functions of variable MICs and composite exposure of drug regimens. Dose increase, if adequately supported by safety studies, likely carries a modest barrier to implementation.

Develop evidence-based patient-stratification approaches

To reliably capture ETT and HTT disease phenotypes, a biomarker signature that integrates host and bacterial factors, and can be converted into a predictive biomarker for clinical practice, is needed. While useful, mycobacterial sputum culture comes with inherent limitations[48], but multiple prospects are emerging for better estimates of bacterial burden than achieved through sputum smear and culture status[87]. Indeed, the field is rapidly evolving toward a composite marker signature that overcomes current limitations[88] and serves as a portable diagnostic tool. Promising avenues include immunodetection of lipoarabinomannan (a cell wall component of Mtb) in sputum and urine[89], GeneXpert (a nucleic acid amplification test that detects Mtb DNA in sputum and rifampin resistance in about 2 h)[90] and peripheral blood transcriptomics signatures of disease extent and TB progression, including incipient TB[91]. New quantitative lateral flow assays have been developed and are being evaluated to quantify host markers of disease severity and/or response to treatment[92]. In clinical trial settings, PET/CT can provide superior and high-resolution pathology evaluation[19] to inform AI-enabled interpretation of chest X-ray in programmatic settings in the future.

Small increases in morbidity and/or mortality associated with excessive treatment shortening or stratified treatment strategies can undermine their benefits. While the 'stop-treatment-and-watch' paradigm has been successfully implemented in clinical trial settings[93], observing patients after treatment completion and initiating re-treatment if needed comes with limited operational feasibility in many geographies. Therefore, the stratification algorithm should be biased toward overtreating 'borderline' patients. To further mitigate the risk of undertreating due to misclassification of patients with ETT disease, leading to relapse and transmission (particularly in high-burden settings)[94], biomarkers capable of evaluating response and risk of relapse in real time are critical[95]. Such markers could also be applied to stop treatment earlier than planned in patients with borderline HTT disease.

We also note that a treatment algorithm that ignores asymptomatic but culture-positive patients with radiological abnormalities (for example, subclinical TB, the 'walking well' or state IV[11]) would fall short of our objective to accelerate the decline of the epidemic. Recent active case-finding studies have revealed higher-than-expected fractions of participants (up to 58% in a high-burden setting[96]) with an abnormal chest X-ray and bacteriologically confirmed TB but no other symptoms; therefore, this substantive patient population likely represents an important source of transmission.

Thus, three categories of biomarkers would enable an improved treatment algorithm: First, a composite signature to distinguish ETT and HTT at baseline and guide regimen composition; second, biomarkers of treatment response to determine when to safely stop treatment and increase confidence in ETT/HTT stratification; and third, a signature for active finding of subclinical TB cases. Clinical trials designed to assess optimized regimens in ETT and HTT cohorts should aim for drastic treatment shortening under the 'stop-treatment-and-watch' principle[93]. Only a dramatic treatment shortening will overcome implementation hurdles in programmatic settings. Table 3 summarizes the research needs, working hypotheses and proposed approaches to enable optimal treatment duration in patient strata.

Conclusion

The goal of TB research organizations and collaborations—including the TB Drug Accelerator (TBDA) and the WHO—is to discover and develop markedly shorter 'pan-TB' treatment regimens[3]. In the ideal world, drug regimen, doses and treatment duration would be optimized for each patient on the basis of an algorithm that integrates host and pathogen factors. Recognizing the realities of TB control, we propose a series of short-term and long-term strategies, from least to most complex, and highlight implementation trade-offs and challenges. We provide pointers to the research community to fill critical gaps in our fundamental knowledge of bacterial vulnerabilities in ETT and HTT disease. We also share our vision of preclinical, computational and translational approaches to build target product profiles and drug regimens with the best potential to cure each disease phenotype. As bioassay technologies improve and AI capabilities grow, the barriers to implementing an accurate stratification algorithm that minimizes overtreating while avoiding undertreatment will lessen—but such progress relies on continued funding investment.

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Acknowledgements

We are grateful to G. Meintjes, M. Hatherill and D. Warner from the University of Cape Town for thoroughly reviewing the manuscript and providing constructive suggestions. We express our gratitude to all members of the TBDA (https://www.tbdrugaccelerator.org/) for their research input, thoughts and fruitful discussions that helped to define the scope and direction of this Perspective, which reflects the views of the TBDA membership. The listed authors are members who directly contributed to the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence should be addressed to Véronique A. Dartois or Clifton E. Barry.

Peer review information *Nature Medicine* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editor: Karen O'Leary, in collaboration with the *Nature Medicine* team.

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